3-Axis Contact Force Fingertip Sensor Based on Hall Effect Sensor

Mirko Raković(✉), Miroslav Beronja, Aleksandar Batinica, Milutin Nikolić, and Branislav Borovac

Faculty of Technical Sciences, University of Novi Sad,
Trg Dositeja Obradovića, 21000 Novi Sad, Serbia
{rakovicm,batinica,milutinn,borovac}@uns.ac.rs,
mickochelsea1234@gmail.com

Abstract. The paper describes the prototype of a novel embedded 3-axis force sensor which is intended to be used for detecting and measuring the contact forces at the fingertips of a robotic hand when grasping and manipulating objects. The sensor is composed of three main parts: a printed circuit board with Hall effect sensors, a neodymium magnet and a elastic silicon layer. The dimensions of the sensor that should be placed at the fingertip are minimized to fit the size of a human inspired robotic hand. The signal processing of the data obtained from the Hall effect sensors is completely done with an ARM Cortex-M4 micro-controller with implemented neural network. The target data which is used for training the neural network is obtained from reference precise 6 axis force/torque sensor. The experimental setup as well as the procedure for acquiring the training data set, learning and implementing the neural network on embedded platform are presented.

Keywords: Robotic hand · Hall effect sensor · 3-axis force sensing

1 Introduction

The humanoid robotics today is continuously expanding toward the goal of efficient operation in complex human centric environments such as offices, homes, schools and hospitals. The human working environment is dynamic and unstructured, thus, to coexist with humans the robot should act in a way that is efficient similar to humans. To achieve this basic requirements, these humanoid robots have to be equipped with many versatile sensors to confer them sufficient autonomy sufficient to perform tasks in such environment. One of the essential tasks exposed to humanoid robots is dexterous manipulation of grasped objects. To make robots capable to do the same, two basic facts have to be kept in mind. The first is that the objects in human's surrounding are mainly designed for humans. The second is that, when manipulating these objects humans are heavily relying on the feedback from tactile sensors. The same should happen for the robots. The tactile sensory system is of paramount importance, providing information to the robot about physical interaction with the surrounding objects.

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The requirements imposed to tactile sensor can be found in [13,16] where tactile sensors are defined as a devices that can measure a given property of an object or can measure continuously variable contact forces. Out of many properties that can be measured, 3-axis contact forces are of the greatest importance because this is essential in grasping and can be used as force feedback in control loop. The papers [8,19] give an excellent review of the current state-of-the-art in different designs of tactile sensors and the analytical comparison between them. Furthermore, paper [12] reviews the techniques for handling tactile data in robotic manipulation covering approaches and applications of tactile sensors in the control of multi-fingered robotic hands.

In [9] a set of sensor design parameters was introduced, making a list which is still widely used by researchers today. The prototypes of tactile sensors for contact force measurement are utilizing different technologies such as resistive, piezoelectric, capacitive, magnetic and optoelectronic. In [4] a cylindrical sensor that can measure three-dimensional applied force is proposed. The main components of the sensor are three coils and a dielectric medium. In [15] the overall design of the Sandia Hand which is equipped with a variety of proprioceptive and exteroceptive sensors is presented. In Sandia Hand, the finger pads can sense the normal contact forces by measuring the deflection of the soft inner layer. In [10] a fingertip for the hands of the iCub robot, where each fingertip consists of 12 sensors is presented. The sensing elements of the fingertip are based on change of capacitance. They are made from a flexible PCB, and a multi-layer fabric that includes the dielectric material and the conductive layer. Authors in [18] designed a flexible and multilayer capacitive microfluidic normal force sensor with a 5×5 sensing elements grid. The sensor uses liquid metal-filled microfluidic channels as capacitive plates and conductive interconnects. In [17] authors have proposed a dome shaped sensor. For measuring the displacement of the dome centre three methods are considered: magnetic, optical and pressure sensors. The authors of [11] designed the sensor that consists of a silicone body in which a small magnet is immersed and a Hall-effect sensor placed below the silicone body can measure the magnetic field generated by the magnet. This sensor can be used to estimate 1-axis normal force applied to the silicone body. The authors in [6] proposed a tactile sensor based on the use of LED-phototransistor couples and a deformable elastic layer.

In [7] the design of BioTac sensor, which consists of a rigid core surrounded by an elastic skin filled with a liquid is described. The skin of the BioTac possesses texture and thickness similar to the properties of human skin. This sensor is capable of detecting cutaneous sensory information that human fingers can detect, i.e. forces, microvibrations, and temperature [7]. The BioTac sensor can be mounted on different robotic hands, as illustrated in [14]. Besides the BioTac fingertip sensor, other tactile sensors for robotic applications can be found such as 3D Force Sensor from OptoForce [1], the FlexiForce from Tekscan [3], and QTC from Peratech [2]. However, the price of these sensors is still relatively high and the performances are not always suitable for the desired implementation. Since the tactile sensor is usually an integrated part of the robotic finger it is not easy to use the existing sensors, and the only possibility is to develop a particular solution.

In this paper the design, development and realization of a tactile sensor that can measure 3 components of the applied force is described. The base of the sensor is a PCB with 4 Hall effect sensors interfaced with ARM Cortex M4 controller that has a 12-bit AD converter. Above the Hall effect sensors the neodymium magnet is placed with the silicon layer in between. The applied force pushes the magnet whose displacement has been recorded by Hall sensors. The complete sensor was placed on top of precise 6 axis force torque sensor used to measure the applied force. To find the relationship between the output of Hall effect sensor and the output of force/torque sensor, i.e. to estimate the 3-axis force based on the output from 4 Hall effect sensors, a neural network (NN) is utilized. The trained NN is implemented in ARM Cortex M4 controller which gives a embedded solution for measuring 3-axis contact force at the fingertip.

2 Structure of the Sensor

The basic measuring element of the proposed 3-axis force sensor is a Hall effect sensor. The Hall effect sensor varies its output voltage in response to the intensity and direction of a magnetic field. The first examples of using Hall-effect sensor for tactile sensing can be found in [5]. Authors in [11] have shown that this concept for measuring the contact force gives high sensitivity, low hysteresis and good repeatability. To measure 3-axis contact force, we used 4 Hall effect sensors and placed them on a PCB as shown in the Fig. 1. The distance between sensors in both x and y direction is approximately 3.2 mm.

As a source of magnetic field, one neodymium disc magnet of 3 mm in radius and 1 mm thickness is positioned above the centre of the PCB. The positioning of the magnet is done manually, but care is taken that the distance from all four Hall sensors is equal. As a consequence, the output voltage of the Hall effect sensor varies with respect to the displacement of the magnet. Since the contact force has to be measured, the distance between the magnet and the Hall sensor should correspond to the intensity of applied force. Thus, a space between magnet and Hall effect sensors is filled with elastic material, i.e. silicone (see Fig. 2). This will allow the displacement of the magnet with respect to sensors once the external force is applied.

An external force F_{ext} applied to the deformable body displaces the magnet from its initial position causing the change of the output voltage of the Hall

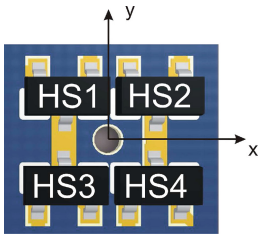


Fig. 1. Position of the Hall effect sensors HS1, HS2, HS3 and HS4 on the PCB board

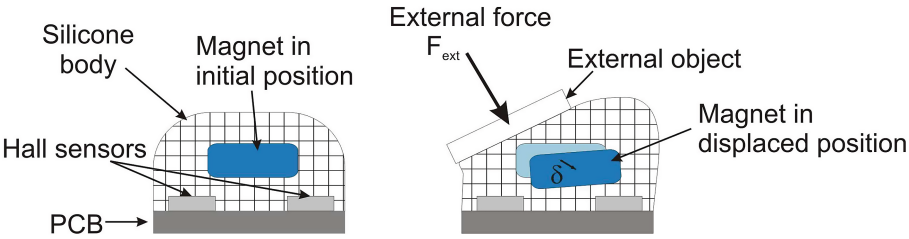


Fig. 2. Working principle of 3-axis sensor

effect sensors. In particular, these deformations produce the change in direction and intensity of the magnetic field with respect to a coordinate frame fixed to the PCB. The deformations can be such as to decrease or increase the magnetic field sensed by the Hall effect sensor.

3 Sensor Characterization as 3-DOF Force Sensor

The magnet motion is not restricted in any direction. It is free floating in silicone body and can rotate or translate. The relationship between the applied force F_{ext} and the 6 degrees of freedom displacement δ as well as the output voltage of the Hall effect sensor against the displacement of the magnet are not linear. This means that the estimation of the external force based on the output voltage of the Hall effect sensors is hard to be determined analytically. For this purpose either a lookup table or some nonlinear regression tool are of great help. For the proof of concept, in this research a classical two layered feed forward neural network is used. In order to collect reference data paired with the output voltage of the Hall effect sensors, the data from precise 6 axis force torque sensor is acquired. For collecting reference data two approaches can be used. The first is to use 6 axis F/T sensor to apply external force. In this case the force intensity is known, but to define its components with respect to the tactile sensor coordinate system is not easy. The second possibility is to place the tactile sensor on top of the 6 axis F/T sensor. If a force is applied it will deform silicon body and will be simultaneously measured by the 6 axis F/T sensor below. In this research, the second approach is adopted. The setup for collecting the paired output from the Hall effect sensors and 6 axis F/T sensor is shown in Fig. 3.

A layer of silicone rubber with magnet on top has been placed on the PCB with mounted Halls sensors. During the data gathering, a plastic stick is used to press the magnet and apply the force manually. The synchronized unfiltered samples from the 6 axis F/T sensor and Hall effect sensors are taken every 10 ms.

As it can be seen from the Fig. 4, the force measured at the reference sensor is in the range of approximately 3 N in x and y direction and 12 N in z direction. The direction and intensity of the force applied to the magnet were varied manually. In total of 6000 samples of data were collected, out of which 70% were used for training and 30% for testing and validation.

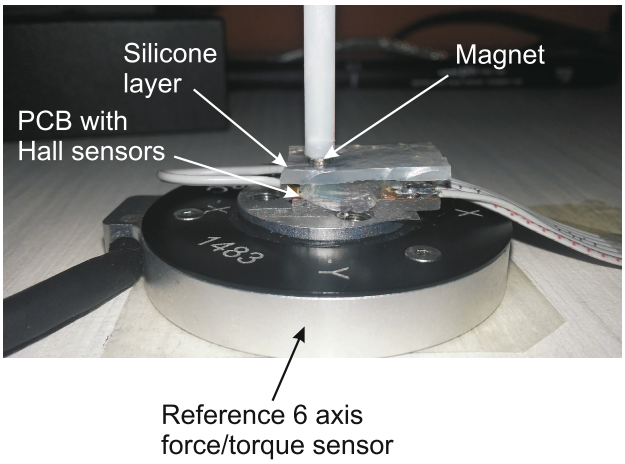


Fig. 3. Setup for collecting the paired output from the Hall effect sensors and 6 axis F/T sensor

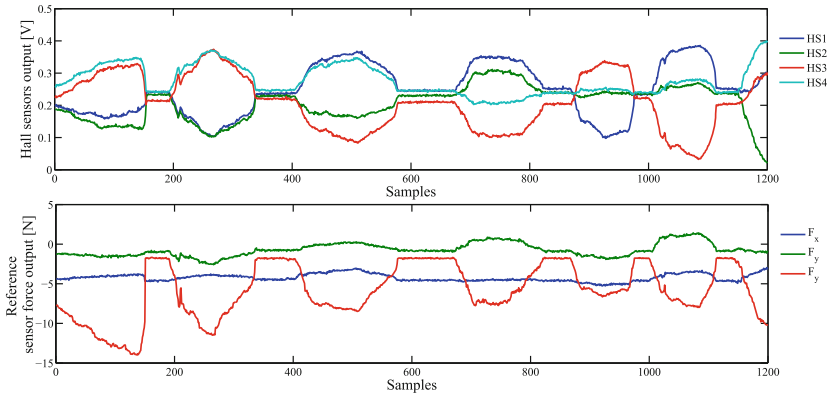


Fig. 4. Samples of paired output from 4 Hall effect sensors HS1, HS2, HS3 and HS4, and three components of the force from the reference sensor

To find the unknown relationship between the outputs from Hall effect sensors and reference force, a feed-forward 2-layer neural network was used. The transfer function of hidden and output layer was set to tangent sigmoid and linear function respectively. For training the neural network, Levenberg-Marquardt backpropagation algorithm was used. Figure 5 shows the comparison between the measured forces from reference sensor and the output from the learned neural network when the data used during training, testing and validation is provided as input.

The obtained NN model was implemented in C code for ARM Cortex-M4 controller, which is the same controller used to read the values from Hall effect sensors. The values from Hall effect sensors are set as input to implemented neural network, thus providing the estimated 3-axis force as output.

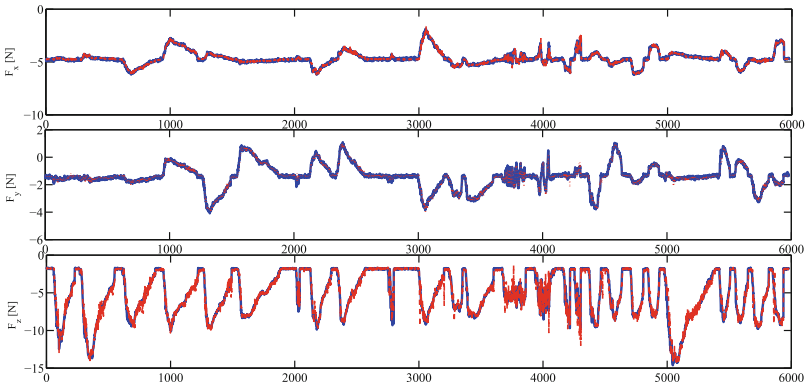


Fig. 5. Compared forces measured by reference sensor (blue lines) and estimated forces obtained by neural network (red lines) when the data used for training, testing and validation is applied as input

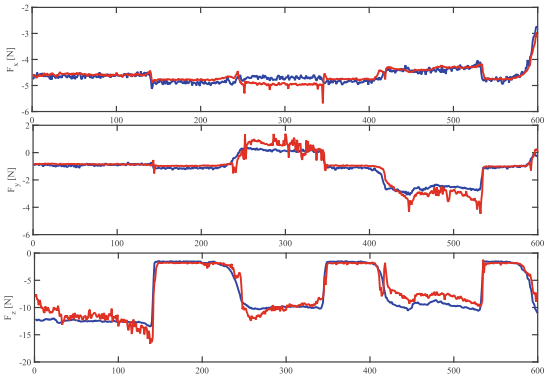


Fig. 6. Compared forces measured by reference sensor (blue lines) and estimated forces obtained by an embedded neural network (red lines) when previously unknown data is applied as input

The estimated output forces and the forces measured from the reference sensor are collected. Figure 6 compares between the estimated forces obtained from embedded neural network model, and the applied forces. It can be seen that there are certain unwanted oscillations of the forces estimated from embedded neural network. The reasons for this can be the over-fitting of the neural network, an insufficiently dense training data set, noisy training data etc.

4 Conclusion

The paper described the concept, realization and experimental characterization of a 3-axis force sensor based on Hall-sensors suited for use as a integral part

of a robotic finger. The mechanical structure of the sensor has been designed so as to allow the device to work as a 3-axis force sensor. As reference sensor a precise 6 axis force/torque sensor is used. To establish the nonlinear relationship between the output from the 4 Hall effect sensors and the output from the reference sensor the neural network is modeled and trained. The training of the NN model is carried out in MATLAB, and after the validation process the final neural network with the identical structure is implemented in ARM Cortex M4 controller. This allowed an embedded and relatively cheap solution for measuring the 3-axis contact force.

Further development will be multi-fold. We will work on the integration of the sensor into the fingertips of a robotic hand in order to use in real-time feedback control systems. More focus will be put on the process of data gathering and filtering the input data. Also, other methodologies for force estimation will be considered.

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